A Clinician-Guided Synthetic Benchmark for AI Detection of Stigmatizing Language in Substance Use Disorder Care

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INTRODUCTION

- Stigmatizing documentation can perpetuate bias and affect care for patients with substance use disorders.[1]
- Existing stigma-detection approaches often rely on keyword lists or annotated clinical notes with restricted access, while attribution, hedging, and clinical context complicate detection.[2-5]
- We developed a clinician-adjudicated synthetic benchmark for AI evaluation, containing no protected health information, and evaluated locally deployed models using zero-shot, rubric-guided, and few-shot prompting.

METHODS

PIPELINE



SYNTHETIC BENCHMARK DESIGN

500 synthetic clinical sentences were generated: 200 clearly stigmatizing, 200 clearly neutral, and 100 ambiguous "gray" cases designed to evaluate contextual ambiguity beyond keyword matching.

Example: "Patient claims her living situation is stable, though she was unable to provide a current address."

Does 'claims' reflect objective clinical documentation or disbelief here? Expert adjudication established the final benchmark label.

CLINICIAN ADJUDICATION

Sentences were rubric-adjudicated by a board-certified Addiction Medicine physician, then independently labeled by a DNP-prepared stigma expert blinded to those labels. Final labels: 251 stigmatizing, 249 neutral. Agreement: 98.8% overall (κ=0.976); 94.0% in ambiguous cases (κ=0.880).

Term	Stigmatizing When
Refused	Frames reasonable behavior as obstinance
Claims	Implies clinician disbelief
Despite	Judgmental framing rather than chronology
Clean/Dirty	Frames test results using moralized language
Attribution	Clinician judgment; not a patient/family quote

MODEL EVALUATION

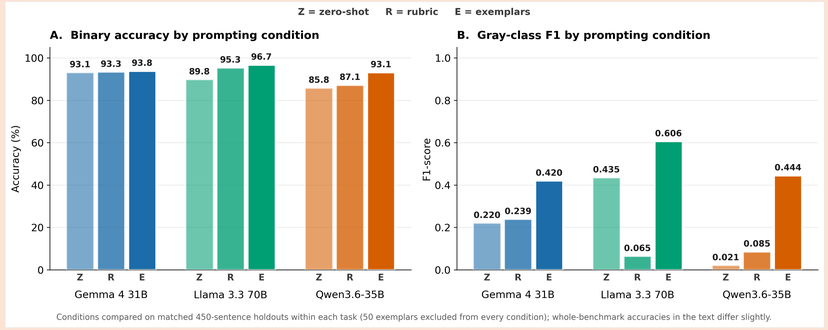
Three open-weight models (Gemma 4 31B, Llama 3.3 70B, Qwen3.6 35B) underwent binary and three-class classification locally on Apple Silicon via MLX under zero-shot, rubric-guided, and few-shot conditions. Conditions were compared on matched 450-sentence holdouts within each task, excluding 50 task-specific exemplars.

EXTERNAL EVALUATION

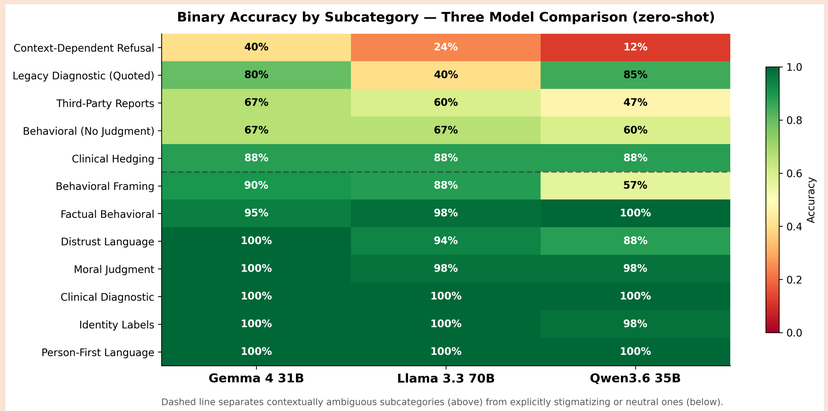
Gemma was evaluated against 3,716 independently annotated examples from the Harrigian MIMIC-IV corpus.[6,7]

RESULTS

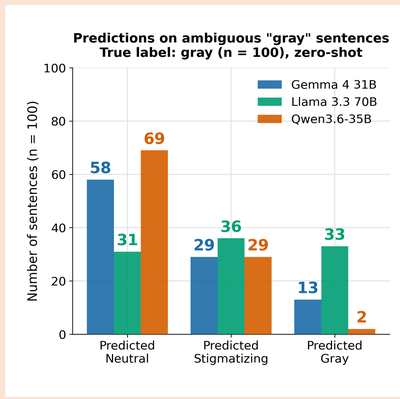
Clinician-labeled exemplars produced the highest binary accuracy and gray-class F1 across all three models. Rubric guidance alone had model-dependent effects: Llama's binary accuracy increased from 89.8% to 95.3%, while gray-class F1 fell from 0.435 to 0.065. With exemplars, Llama achieved 96.7% accuracy and gray-class F1 of 0.606.

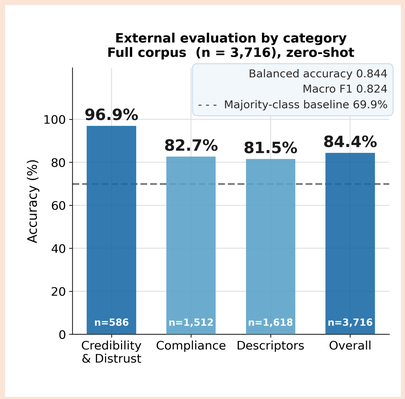


All models achieved 100% accuracy on person-first and clinical diagnostic language. Performance varied across categories and was lowest on context-dependent refusal.



Under zero-shot prompting, Llama assigned the gray class more frequently than Gemma or Qwen. In external evaluation, Gemma achieved 84.4% accuracy and 84.4% balanced accuracy on 3,716 independently annotated MIMIC-IV examples. Accuracy exceeded the 69.9% majority-class baseline. Stigmatizing-class F1 was 0.883.





CONCLUSION

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- Clinician-guided synthetic benchmarks support evaluation of AI for stigmatizing-language detection without exposing protected health information.
- Strong overall accuracy concealed difficulty recognizing contextually ambiguous language.
- Clinician-labeled exemplars improved performance across all three models, while rubric guidance alone had distinctly model-dependent effects.
- Gemma achieved 84.4% accuracy on 3,716 annotated clinical examples, exceeding the 69.9% majority-class baseline.
- Local inference on consumer Apple Silicon makes evaluation feasible without external data transmission. Further clinical evaluation is needed before use in addiction care.

LIMITATIONS

- Synthetic sentences, generated by a single language model, may not fully reproduce the complexity or longitudinal context of clinical notes.
- High inter-rater agreement reflects two reviewers applying a shared rubric and does not establish a universal definition of stigma.
- Findings may vary across models, prompting strategies, and exemplar selection.
- Effects on clinician behavior, documentation quality, and patient outcomes were not assessed.

AUTHORS & DISCLOSURES

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